AK-07 Project Log: Day-by-Day Technical Record

Oct 19, 2026 · @Olwethu

About this log

The work ran from the PCB stator files in September to rev. C of the predicted operating points on 19 October 2026. This log records each working day: what was done, the numbers obtained, what failed and how it was fixed.

How to use it. It is source material for the report's methodology, results and "project management" sections, and for the AI-use declaration. Each entry names the model or run (project name, profile, HPC job number) so the claims can be traced to files in the GitHub package.

Sources. The project documents saved during the work, the run logs and the result files. Dates before 4 October are less precise: the September work is recorded by topic rather than by day, so it is grouped.

Conventions used throughout.

- K is the torque constant, defined by $T = 1.5 \cdot K \cdot \hat{I}$, where $\hat{I}$ is the peak phase current. It is in mN·m/A, which is numerically equal to mWb/rad, so the back-EMF is $e = K \cdot \omega_m$.
- Board 1 is the board nearest the magnets.
- 1000 rpm = 133.3 Hz electrical (8 pole pairs).
- "Lab PC" means the 16 GB, roughly 4-core UCT lab machines (RedLab17/18/19, GreenLab16).

Before 4 October: stator files, fabrication and the first Maxwell model

The stator came from the open-source gouldpa00/magnetless_motor design (GPL-3.0). The September work turned it into a board that could be ordered and modelled. These items are grouped by topic, because the September conversations were not logged by day.

Mid-September (by 14 Sep): reading and ordering the PCB.

- The Altium Gerbers and the PcbDoc were inspected in Python, so that board parameters were confirmed from the files rather than from the design notes:
  - `pcb-tools`, patched for Python 3.12 by replacing the `'rU'` file mode in three modules;
  - `olefile` to read the PcbDoc;
  - `GerberCairoContext` to render the layers.
- **Confirmed board:** the "mini_wave-bb" variant. It is a 12-sided board, 93 mm across flats, with a 25 mm bore. It has 6 layers of 1 oz copper and a three-phase wave winding with 16 coils per phase, for 16 magnets (20 × 6 × 2 mm). It carries 617 plated vias (0.3 mm drill, 0.5 mm pad) and 12 non-plated 3.1 mm holes.
- **Ordering:** a cleaned Gerber zip and a field-by-field JLCPCB ordering guide were prepared. The guide stressed three settings: inner copper set manually to 1 oz (the default is 0.5 oz), ENIG finish, and 1.0 mm board thickness as a deliberate air-gap decision. One flag was raised: the JLCPCB link in use pointed to the stencil flow, not PCB fabrication.
- A single mini board was driven by an ESC under a magnet rotor and spun, confirming the winding works.

Late September: Gerber import into Maxwell, which was abandoned.

- Importing the six copper layers made Maxwell split the PCB into about 430 disconnected conduction paths.
- Equivalent copper connectors through the missing via sections fixed the connectivity. The result was three continuous phases, each of 164 copper objects, with matched copper volume (0.001 % difference) and no cross-phase contact.
- The excitation then failed. The large solder-pad faces (about 52 mm² on the top layer) are not valid coil-terminal cross-sections, and two faces in one terminal triggered "total number of conductors should equal to one".
- **Decisions at this point:**
  - build the stator natively in Maxwell from the PCB source files, by a one-click AEDT Python script, rather than import it;
  - fix the Rev0 specification: 5 boards of 1.0 mm in series per phase, connected in star, with one N35 rotor on an 86 mm OD, 2 mm steel backplate (19 mm bore) at a 1.0 mm air gap;
  - work with Olwethu defining the motor and Claude implementing the model.
- The lab PCs were noted as the binding constraint: 16 GB RAM and about 4 cores.

3 October: native build script BUILD_PCB_STATOR_MAXWELL.py.

- **What the script builds:**
  - copper per layer and per net read from the ODB++ data (PCB_MAXWELL_SOURCE_PACKAGE), each feature buffered to its real width and unioned per net;
  - 25/25/25/24/24/25 islands per phase on L1–L6, which is 148 per phase;
  - 615 unique vias as 0.30 mm solid copper.
- **Checks:** each phase forms exactly one connected component, and the minimum spacing between phases is 0.241 mm. Layers L2/L3 and L4/L5 carry identical copper in parallel, so each phase has internal loops.
- **First full transient run:** the build took 19 min and validated. Meshing gave 1.93 M tetrahedra in 32 min, using 10.4 GB. The solve failed at step 1 with "Error in calculating the conduction paths".
- **Fix:** raised straight leads (1 × 0.5 mm) from the L1 start pad and the L6 end pad, running to the region's +Y face. The coil terminals are the lead end faces.

4–7 October: one board validated, geometry fixed, the full motor too big for the lab PCs

By the night of 7 October the single-board model was validated against eight separate checks, the air gap (1.0 mm) and backplate (2.0 mm) were fixed, and a five-board frozen build existed. However, the complete motor could not be solved on 16 GB.

Saturday 4 October

- **Rev0 specification frozen** as the single source of truth, with PCB_MAXWELL_SOURCE_PACKAGE. Pipeline REV0_PIPELINE v5.2 was built with checkpointed stages: project → stator → rotor → region/band → excitation → solver → validate → analyse → export.
- **Acceptance checks** were built into every run: the reports must exist, there must be no new solver errors, exports must be non-empty, and the solved current must match the source within 5 %.
- **Gate 1 (magnetostatic, 1 A DC, one phase at a time): PASSED.** The resistance from the ohmic-loss integral was A 0.27776, B 0.27736 and C 0.27789 Ω. That is a 0.19 % spread and +2.9 % against the 0.2699 Ω design value.
  - Each phase is 148 islands and 205 vias, with In and Out 25 island-hops apart.

- Peak current density was 3.8×10^{7} A/m² at 1 A.

Sunday 5 October

- **Gate 2A (stationary T-Ω transient, Phase A): PASSED** (project REV0_GATE2_TOMEGA_PHASEA_V5).
  - The solved current was 70 mA against 69.76 mA requested.
  - The flux linkage gave L ≈ 2.08 µH and the induced voltage (1.7313 mV) gave 2.07 µH, agreeing within 1 %.
  - The PC froze at "Start Matrix", but the results survived.
  - An earlier v4 run had solved with 0 A because the variables were created after the windings; this was fixed in v5.
- **Board three-phase magnetostatic (FR-4 and mask included):** froze at pass 3 "Start Matrix" after 2 h 20 min (solver at 8 GB). The last adaptive pass was recovered with postprocess_ms.py: R within 0.5 % of Gate 1, and no cross-phase current. The inductance matrix was turned off by default from then on.
- **FR-4 and solder mask modelled as air:** approved by Olwethu and verified, changing R and magnetic energy by ⩽ 0.22 %. Precedent: Han et al. (ECCE 2021) model only the copper of PCB coils.
- **REF_ROTOR_MS_SMOKE (rotor + board 1, 0° and 11.25°, ±1 A): PASSED.** Four solves of about 20 min each.
  - The rotor virtual-work torque and the copper Lorentz torque agreed within 0.05 %.
  - The ±I average exposed a −0.47 mN·m virtual-torque bias, which is numerical. Copper torque was used from then on.
  - A first estimate gave K1 = 5.81 mN·m/A per board.
- **Reviewer corrections (v6.1):**
  - sweep a full electrical period (0–45° mech in 5.625° steps);
  - check periodicity and antisymmetry;
  - cap the gap mesh at 300 k elements;
  - a locked sequence: baseline → three-phase check → DOE → re-check → five boards summed at their actual positions, never "×5".
- **Phase geometry check in Python:** Phase B = Phase A rotated −30° mech, and C = A rotated −60° mech, with ⩾ 99.96 % copper overlap on every layer. This confirms 0 / −120 / +120° electrical.

- **22:21: the lab PC crashed** during the 18-solve sweep. Optimetrics saves only at the end, so the run was lost. v6.6 added a stepwise sweep that saves after every angle and resumes.

Monday 6 October

- **REF_ROTOR_MS_SWEEP baseline: PASSED at 05:05**, resumed after the crash. The angles 28.125–45° were solved from 02:09 to 04:57.
  - K1 = 6.09 mN·m/A per board-phase, ψ = −30.2° electrical.
  - λ1 = 0.7615 mWb peak, giving 0.451 V rms per board-phase at 1000 rpm.
  - Full-period repeat 0.02 %, half-wave antisymmetry 0.05 %, rotor against copper torque ⩽ 0.12 %.
  - H3/H1 = 11.75 %.
  - Cogging was below the ±0.5 mN·m numerical noise.
  - Steel peak |B| was 1.39–1.44 T; the mid-gap mean |Bz| was 0.168 T.
- **Inter-board connection approved:** a 1 × 1 mm copper rim "solder bridge" from board k L6 Out to board k+1 L1 In, with raised leads on the string ends only and no star point yet.
- **First five-board build (v7):** used one united solid per phase. It ran for more than 2.5 h and AEDT closed. The fix came in v7.1–7.3, following Olwethu's workflow:
  - no cross-board union;
  - history purged per board;
  - each board built, validated, saved and checkpointed before the next.

Tuesday 7 October

- **REV0_STATOR built:** all 5 boards checkpointed, with 12/12 junction probes passing. The first run failed with "illegal external terminal" because the +Y padding was 5 mm, not 0. This was fixed in v7.5, and Olwethu froze REV0_STATOR_FROZEN.aedt.
- **STACK_JUNCTION_TEST (2 boards, DC): PASSED.** The string resistance was 0.55313 Ω against 0.5554 Ω expected (−0.4 %), with 3.6 µW in the bridge. L_string was about 6.23 µH, so M12 ≈ 1.04 µH.
- **Design-of-experiments points** (single board, 5 mm padding):
  - 0.5 mm: K1 = 7.11;
  - 5.0 mm: K1 = 1.815;
  - a 5 mm backplate: K1 6.097, only +0.1 % over 2 mm.

  - The three-point decay fit gave k = 0.3031 /mm (residual ⩽ 0.17 %), and the stack sum gave 18.05 mN·m/A.
- **16:15, geometry locked by Olwethu:** g = 1.0 mm, t = 2.0 mm. A 5 mm plate would add about 130 g for +0.1 %, and 1.0 mm leaves clearance for board bow, run-out and bearing play.
- **REV0_FINAL_G100_T20_FROZEN** was built, with 36/36 junction probes passing.
- **Three-phase superposition check: PASS** (REV0_CHECK_PHB on RedLab17, REV0_CHECK_PHC on RedLab18).
  - B: K1 +0.31 %, ψ +0.20° off ideal.
  - C: K1 +0.03 %, ψ +0.40° off ideal.
  - Mean synchronous torque 9.1478 against 9.1375 mN·m/A pk ideal (+0.11 %).
  - The reversed sequence gives about 0, so the phase sequence and polarity are confirmed.
  - The direct one-board three-phase run (REF_ROTOR_MS_3PH) crashed RedLab17 at 8.64 GB.
- **Three full five-board solve attempts on 16 GB: all failed.**
  - RedLab18 froze at 18:08.
  - RedLab18 froze again at about 19:00 during initial meshing (mesher at 5.5 GB).
  - RedLab19: the validation check alone took 530 s, and the solve started with only 4.03 GB free. AEDT closed after 1 h 37 min, at 23:02.
  - **Conclusion:** a documented hardware limit. The board-by-board method was adopted, and an HPC run or a 64 GB workstation was recommended.
- **22:34: MATLAB lumped model, Rev0 final.**
  - e~x~ = ω·k~x~(θ), J·dω/dt = T − T~L~ − B·ω, with R = 1.389 Ω and L ≈ 25 µH.
  - Result: 27.0 mN·m/A pk and 2.32 V rms line at 1000 rpm.
  - Torque ripple at 1 A: 0.70 % with sine drive, 14.4 % with six-step.
  - On 12 V: 3671 rpm at no load.
- **22:56: Rev1 split into two designs** (the reviewer's proposal): Rev1-D (S1–[PM|steel|PM]–S2–[PM|steel], 48 magnets) and Rev1-S (S1–R1–S2–R2, 32 magnets).
- **From 23:23** the Rev1 board runs were started on three PCs, and REV1D_FINAL was built (23:28 → 00:16; 48 magnets; S1 and S2 each 36/36 probes).

8–12 October: Rev1 boards, the field map, thermal and the complete Part C board set

By 12 October every board position of Rev0, Rev1-S and Rev1-D had a detailed FEM result with 20 mm padding. The whole-motor field map was validated against it (+1.4 % at board 1), and the thermal and system models existed.

Wednesday 8 October

- **Rev1 detailed boards** (5 mm padding):
  - **REV1D_S2_K1** (RedLab17; one crash, then a restart from scratch, done about 04:00): K1 8.4860 mN·m/A against 7.8711 predicted by superposition (f = 1.0781).
  - **REV1D_S2_K3** (middle board): K1 7.0504 (f = 1.0726).
  - **REV1_S2ENV** (the Rev1-S S2 board next to R1′s bare steel): K1 2.5970, which is ×1.467 the free value, giving an image coefficient c = 1.166.
  - S2 is in phase with S1 (+0.1°), so S2 is wired in the same sense.
  - The Rev1-D shared 2 mm plate stays unsaturated (≤ 1.30 T).
  - R1 and R2 attract each other with about 47–48 N, a mechanical load for the shaft, spacers and bearings.
- **Stack sums (5 mm basis):** Rev0 18.05, Rev1-S 38.54 and Rev1-D 56.89 mN·m/A.
- **01:10, decision: the full-motor field map,** reported as "field-only FEM with trace-resolved winding post-processing" (v9.5).
  - Maxwell solves only the magnets, steel and air at 8 angles.
  - The real track centre-lines (traces_board.json, from the ODB++ data, with a DC nodal solve giving each track piece′s current) carry 1 A in post-processing. That is 180 775 points for Rev0 and 361 550 for Rev1.
  - Torque: T = −I·B~z~·(x·dl~x~ + y·dl~y~) on tracks, plus I·dz·(x·B~x~ + y·B~y~) on vias.
- **Validation gate initially FAILED (+4 %),** and the root cause was found: the region boundary. With 5 mm padding the B·n = 0 face sits 5 mm below the board and suppresses B~z~. An analytic image model predicted the field-map/detailed ratio as 1.039; 1.041 was observed.
- **05:49, gate PASSED** after rerunning with 20 mm padding: the detailed REV0_DOE_D100_P20 gave K1 6.2816 against field-map board 1 6.3697 (+1.40 %, 0.04°). This was inside the 2 % / 1° limits.
  - Field-map machine K (P20): Rev0 19.50, Rev1-S 41.64, Rev1-D 60.28 mN·m/A.

  - Mesh was converged: 0.8 → 0.55 mm changed the result by 0.03 %.
- **05:36: MATLAB v2 with the field-map K.** On 12 V: Rev0 3393 rpm at no load, 68.3 % copper-limited efficiency at 20 mN·m; Rev1-S 1589 rpm; Rev1-D 1097 rpm, with no headroom at 20 mN·m.
- **Icepak thermal (lab PC)**, with an assumed housing: an acrylic tube in the bench hood, 30 °C air and 1 m/s fans. The rotors were still. Loss was iterated with R(T).
  - **Rev0** (REV0_ICEPAK_STEADY_RUN1, 05:28):
    - 1 A pk gives 2.32 W, with the hottest board at 50.3 °C and R~th~ 8.6 K/W;
    - 2 A pk gives 11.6 W and 124.9 °C (fans on), or 134.5 °C (fans off).
  - **Rev1-D (08:36) and Rev1-S (about 09:20):** at 1 A pk, about 4.85 W and a 65.7–67.1 °C hotspot; S1 runs 2–15 K hotter than S2.
  - **Lumped network** calibrated to Icepak (within ±7 %): the Rev0 continuous limit was 1.75 A pk at a 100 °C board limit. This was later withdrawn on 17 Oct.
  - **Read-out bugs fixed:** "Tmag" was a volume maximum at interface nodes, and v1 converted any value above 200 to Kelvin.
  - **Mesher workaround:** 20 thin stacked bodies made Mesher-HD drop a winding, so each stator was merged into one body.

Thursday 9 October

- **Decision (Olwethu):** no full detailed-motor solve. Three attempts had failed, and the HPC route looked too late. Part C would be completed with detailed single-board FEM at every board position, with 20 mm padding, summed board by board.
- **Part C v2 scripts.** The single-board templates had been lost because the lab PCs wipe C:\Users\Student on sign-out. v2 rebuilds every model from the embedded geometry. Post-processing run_all_post.py stages P1–P8:
  - P1 fits;
  - P2 stack sums against the field map;
  - P3 linearity;
  - P4 cogging;
  - P5 field decay;
  - P6 skin depth and eddy loss;
  - P7 three-phase EMF and ripple;
  - P8 summary.
- **22:55, runs started** on GreenLab16/17/18. There were two fixes that night:

    - 22:40: the folder search had picked an old pipeline ("no module named aedt_api");
    - 00:15: WORK_DIR was forced to C:. The pipeline had used D: whenever it existed, and on GreenLab16 D: was a flash drive that was removed at about 23:55.
- **Rev1-D with 20 mm padding was too heavy for 16 GB.** J1 froze at about 23:15, J5 froze at 01:28, and J3 was rejected at 5.625°. At 01:45 these were switched to 5 mm padding, since S2 changes by only 0.01–0.03 % between 5 and 20 mm.

Friday 10 October

- **Complete Part C board set** (detailed board / field map):

Board	Rev0 S1	Rev1-S S2	Rev1-D S2
1	6.2816 / 6.3697 (−1.4 %)	2.5965 / 2.8842 (−10.0 %)	8.4860 / 8.7539
2	4.5905 / 4.7962 (−4.3 %)	3.0572 / 3.3286	7.4008 / 7.8575
3	3.3655 / 3.5881 (−6.2 %)	3.8115 / 4.0853	7.0504 / 7.5412
4	2.4755 / 2.7268 (−9.2 %)	4.9382 / 5.1866	7.3975 / 7.8497
5	1.8266 / 2.0154 (−9.4 %)	6.5593 / 6.6506 (−1.4 %)	8.4791 / 8.7587

- **Machine K (detailed sum):** Rev0 18.54, Rev1-S 39.50 and Rev1-D 57.35 mN·m/A. The field map is consistently about 5 % higher.
- **Checks:**
    - ψ = −30.0 to −30.2° on every board.
    - Rev1-D mirror symmetry: 0.08 %.
    - Linearity, 2 A against 1 A: −0.002 %.
    - Cogging below noise (0.47 mN·m, not periodic).
- **P5–P6:**
    - Fundamental B~z~ on the board-1 tracks was 0.174 T, and on board 5 0.051 T; the decay was 0.300 /mm.
    - Skin depth was 5.72 mm at 1000 rpm, so R~ac~ ≈ R~dc~.
    - Thin-strip track eddy loss at 1000 rpm: Rev0 19.6, Rev1-S 42.7 and Rev1-D 87.8 mW.
- **Finding: the method gap grows with distance from the magnets** (+1.4 % at board 1, up to +10 % at board 5). The boundary was ruled out (both models used 20 mm), and so was the mesh (≤ 0.1 %). The shaft geometry was checked on 12 Oct. The gap was

attributed to the method itself: centre-line filaments against volume force.

Saturday 11 October

- **HPC account application submitted** (awaiting Prof Khan), reversing the 9 Oct decision. The HPC reference values were frozen as the acceptance basis: copper against rotor torque within 3 %, and the result flagged if more than 2 % outside [detailed sum, field map].
- **MATLAB v3** (K source switchable between detailed and field map; Octave-checked) at 12 V and 20 mN·m:
  - Rev0 3568 rpm at no load, 66.0 % copper-limited;
  - Rev1-S 1675 rpm, 81.5 %;
  - Rev1-D 1154 rpm, 90.3 %.
- **hpc_motor.py, first versions:** a single self-contained file (about 2 MB) carrying the pipeline, geometry and references. It was tested against a mock AEDT.

Sunday 12 October

- **Round-2 lab runs:**
  - Shaft test REV0_DOE_D304/D508_P20_SH (RedLab18): ⩽ 0.016 % change, so the shaft geometry was ruled out.
  - REV0_H8_D100_P20 (RedLab17, board 1 at 8 angles): K1 6.2887, with H3 10.84 %, H5 0.99 % and H7 0.34 %.
  - The FIELDMAP_*_P20X trace exports reproduced 8 Oct within 0.3 %.
- **MATLAB v3.1** added the 5th and 7th harmonics: torque ripple 0.85–0.88 % with sine drive and 14.8 % with six-step.
- **Priority set:** full rotating transients for Rev0 and Rev1-S; Rev1-D static only.

13–16 October: the HPC full-motor solve settles the method; simplified rotating models

The Rev0 full-motor solve on the UCT HPC (job 1470376) gave K = 18.534 mN·m/A, within 0.03 % of the board-by-board sum. That settled which reduced method to trust. The detailed copper was too slow to mesh for rotation, so two simplified rotating models were built.

Monday 13 October

- **UCT HPC access granted** (user khnolw003): 10 GB home, a 50 GB volatile /scratch, 40 cores per job and about 7.8 GB per core. Ansys module software/ansys-2025R2-v252.
- **hpc_motor.py v5:** one self-contained file of about 2 MB, submitted with `sbatch`. It contains:
  - a supervisor that runs each stage as a separate AEDT batch process and resumes after failure;
  - a 40-core cap;
  - a **disk guard** that stops AEDT before the shared work folder passes 45 GB;
  - the AEDT TMPDIR on node-local disk when available;
  - gates for split-shaft equivalence and the short-step transient;
  - a Rev2-S builder (rev2.py).
- **Priorities set:** 1. Rev0; 2. Rev1-S; 3. Rev2-S if time. Rev1-D was dropped from the HPC plan, and its static results stand.
- **About 23:33: HPC smoke test** (Gate 1 on 8 cores) ran successfully.

Tuesday 14 October

- **02:14, job 1470229 (Rev0, "friday" mode) refused to start.** The account-wide /scratch quota is not readable on the compute nodes, and the script would not run unprotected. It was resubmitted with `--allow-no-quota`, relying on the work-folder guard, as **job 1470376** (02:20; 32 cores, 243.8 GB).
- **Rev0 full-motor magnetostatic, angle 0°** (solve_00): all 5 detailed boards plus the rotor, with three-phase synchronous currents at 1 A pk, iterative solver, 6 adaptive passes. Started 03:12, finished 20:39: **17 h 27 min, 109.7 GB peak**.
  - Copper torque 27.893 mN·m against rotor virtual work 27.749 mN·m (0.51 %).
  - Maxwell kept only about 2.9 of the 32 cores busy on average, so the run was memory-bound rather than parallel.
- **The detailed copper could not be meshed for rotation.** Maxwell's mesher runs on one core:
  - 1 board, 1 phase, transient: about 95 min of meshing;
  - 1 board, 3 phases plus rotor and band: still meshing after 3.6 h at 44 GB, and stopped by the wall time.
- **Decision (Olwethu): two simplified full-motor rotating models (v10),** tried on the lab PCs first:

- **Model A, FMROT (fm_rotating.py):** magnets (eddy currents on), steel and air turning at 1000 rpm in a rotating band (two bands for Rev1-S). B is exported at every step on the 180 775 track points, giving k(θ), back-EMF, the loaded torque with synchronous currents, cogging and copper eddy loss σ/12·(w²·Ḃ~z~² + t²·Ḃ~n~²).
- **Model B, EQ (eqcoil.py):** 8 stranded sector coils per phase per board, measured from traces_board.json (sides at 18.8° and 41.1°, 7.5° wide, r 18.6–36.6 mm, 2.03 geometric turns), with A on L1+L6, B on L2+L5 and C on L3+L4. A calibration run (EQ_*CAL) is followed by rotating runs (EQ*_ROT).
- Ten runners were written, and both models were tested end to end on a mock of the AEDT API.

Wednesday 15 October

- **14:18, Rev0 angle 3.75° done** (17 h 39 min, 109.3 GB). Copper 27.710 against rotor 27.659 mN·m (0.18 %).
- **Rev0 full-motor result:**
  - **K = 18.534 mN·m/A** (copper), 18.470 (rotor virtual work).
  - Back-EMF 1.941 V peak, 1.372 V rms per phase at 1000 rpm.
  - Against the board-by-board sum (18.540): −0.03 %. Against the field map (19.496): −4.93 %.
  - Per board, 6.2817 / 4.5898 / 3.3637 / 2.4736 / 1.8254, each within 0.07 % of the single-board runs.
  - Steel |B| max 0.92 T.
  - **Conclusion:** the board-by-board method is exact for this coreless stator, and the field map is a consistent ~5 % upper bound.
- **09:07, Rev1-S full-motor static, job 1486831** (40 cores, 304.7 GB). The stator build took 41 min, the final assembly 21 min, and solve_00 started at 10:09.
- **14:28: Rev0 short-step transient gate** (mesh MT0) started in job 1470376.

Thursday 16 October

- **Hardware arrived:** all 10 PCBs.
  - The board phase resistance measured 0.32–0.34 Ω (U3401A, 2-wire, relative mode), giving a 1.54–1.83 Ω stator phase against the model's 1.39 Ω.
  - The built backplate is 80 mm OD and 25 mm bore (model: 86 / 19 mm).

- **Decision (Olwethu): Rev1-D dropped;** Rev2-S to replace it if time allows. The hardware plan became Rev0, then Rev1-S.
- **Provisional test specification** (Docs doc, 16 Oct).
  - Operating points at 1000 rpm and 1 A pk, in star.
  - A sensitivity of K to the air gap of about −7 % per 0.25 mm, and about −4 % per 0.1 mm of inter-board gap.
  - A provisional thermal limit of 0.85 A pk per stator (15 K/W, 60 °C board limit). The old 1.75 A pk was withdrawn.
  - L ≈ 110 µH per stator from a Neumann sum. This was **later found to be about 4× too high and corrected to about 25 µH on 19 Oct.**
- **HPC v10 runs:**
  - **FMROT_REV0_SHORT (job 1498566, 16:04–17:08): PASS.** Over 6 steps, the largest point difference against the static field map was 1.33 % of K1 (limit 3 %).
  - **EQ_REV0_CAL (job 1498603): FAILED twice** ("Setup1 has failed with execution error", 16:23 and 16:40).
    - Cause: slivers between adjacent phases' coil slabs (0.26° overlap and 0.05 mm gaps) broke the mesh.
    - Fix: coil sides trimmed by 0.8° and slabs thinned to 0.08 mm.
    - The calibration target was changed to the full-FEM per-board values rather than the field map.

17–19 October: rotating results, disk-quota losses, reports, ratings and publication

Both rotating models passed against the static field map, within 1.49 % (Rev0) and 1.59 % (Rev1-S). Both equivalent-coil models were calibrated. The shared 50 GB scratch quota cost two long runs, including a 48 h Rev1-S full static solve.

Friday 17 October

- **09:03–09:09:** EQ_REV0_CAL (job 1500327), FMROT_REV0 (1500328) and FMROT_REV1S (1500332) were submitted beside the running Rev1-S static job.
- **About 09:30: the disk guard stopped every job at 45 GB.** This included job 1486831, the **Rev1-S full static, after 2 days 0 h 23 min and 250 GB peak memory**. It had not finished its first angle, and it could not be resumed.

    - **Fixes:** clean-up; rotating runs reduced from 48 to 24 steps per period (a saved field frame costs about 0.5 GB per step for Rev0); one large job at a time.
- **10:03: rerun.**
    - **EQ_REV0_CAL (job 1500351, done 13:45): USABLE.**
        - Geometric turns (Ncoil 2.006) gave K ≈ 23.0 mN·m/A, which is 24 % high.
        - Calibrated to the full-motor per-board values: Ncoil 1.623 (× 0.809).
        - Board spread 7.1 % (board 1 about +4 %, board 5 about −3 %), phase spread 1.14 %, ψ error ⩽ 0.4°.
        - The terminal polarity was reversed and is corrected in post-processing.
        - It is a calibrated model, not an independent prediction.
    - **FMROT_REV0 (job 1500352, done 13:06): PASS.**
        - Over 120 board-phase-angle points, the worst difference was 1.49 % of K1. Board 1 was within 0.4 % and 0.2°.
        - K = 19.475 on the field-map basis, scaled × 0.9517 to 18.534. That gives 1.372 V rms phase and 2.377 V rms line at 1000 rpm.
        - THD 5.5 % phase (H3 5.5, H5 0.67, H7 0.05 %) and 0.7 % line.
        - Torque ripple 1.26 % pk-pk at 1 A pk (sinusoidal currents), and cogging 0.61 mN·m pk-pk, not periodic.
        - Track eddy loss 20.4 mW at 1000 rpm (boards 1–5: 10.8 / 4.8 / 2.5 / 1.5 / 0.8 mW).
        - Current density per 1 A: median 9.2–9.4, 95th percentile 10.3, maximum 57.1 A/mm².
    - **FMROT_REV1S (job 1500353): out of memory after 57 min at 16 cores** (113.8 GB). Rev1-S needs 32 cores.
- **11:07:** FMROT_REV1S_SHORT at 32 cores (job 1500415, done 13:05): **PASS, 1.51 %.** EQ_REV1S_CAL started as job 1500416.
- **The FMROT field maps were all NaN.** The export probe accepted a headerless point file in mm that AEDT read as metres. Fixed in hpc_motor sha 0d6cd6f14f1f, which accepts only a variant returning finite B.
    - The post-processing rerun (job 1500700) was refused: `--time=04:00:00` is shorter than lab mode's estimate. Job 1500707, with 24 h, completed.
    - Results: stator-face B~z~ max 0.448 T and backplate |B| max 0.907 T (static: 0.92 T).
- FMROT_REV1S, the full 24-step run, was submitted as job 1500701.
- **Rev0 technical report and figure zip** were delivered: back-EMF, torque and cogging, field maps, current density and efficiency figures, plus the CSVs.

- An external reviewer's critique was incorporated:
  - Kt quoted per A rms too (39.3 mN·m/A rms);
  - the equivalent-coil model called calibrated;
  - the zero eddy loss explained as a consequence of the field-only scope;
  - a ripple cross-check, 2·(H5 − H7) = 1.23 % against 1.26 %;
  - 1.75 A pk withdrawn;
  - stack thickness up to 6 mm would lower K by about 9 %.
- **Decision (Olwethu):** keep the modelled geometry. The report states that the hardware (backplate OD, gaps) shifts accuracy, and recommends building and measuring first, then modelling as built.
- **About 20:00: the disk guard stopped jobs 1500701 (FMROT_REV1S) and 1500416 (EQ_REV1S_CAL).** EQ_REV1S had solved 7 of 8 angles, but the project file was corrupted, so OpenProject failed.

Saturday 18 October

- Job 1502053 failed with "budget used up before the start": it was submitted before the clean-up delete had finished. It was resubmitted alone as **job 1502103** (FMROT_REV1S, 32 cores, 122 GB peak).
- **About 20:00, job 1502103 completed: PASS**, worst 1.59 % over 240 points.
- Olwethu planned to start hardware running on the simulated ratings, then send as-built dimensions for a remodel.

Sunday 19 October

- **Rev1-S results.**
  - **Equivalent-coil model, calibrated offline:** the missing 8th angle (39.375°) was filled by half-wave antisymmetry, T(θ + 22.5°) = −T(θ), which matched to 0.08 mN·m on the solved pairs.
    - Ncoil 1.631 (× 0.813), board spread 9.0 %, phase spread 0.87 %.
    - With Rev0's factor unchanged it gives K = 39.30, against the board-by-board 39.50 (−0.5 %).
  - **Rotating model (series), scaled × 0.950:**
    - 2.925 V rms phase and 5.066 V rms line at 1000 rpm; THD 5.2 % / 0.6 %.
    - 59.25 mN·m at 1 A pk, ripple 1.16 %, cogging 0.78 mN·m pk-pk.
    - Fields: R1 steel 0.835 T, R2 steel 0.931 T; S2 face toward R1 steel 0.108 T.

        - Eddy loss 44.2 mW.
    - The Rev1-S technical report and the first ratings sheet were written, using S1 measured at 1.8 Ω and 0.85 A pk.
- **GitHub package** (pcb-axial-flux-motor-maxwell, about 60 MB at first):
    - branches main (scripts, PCB source package, mock-AEDT test harness, analysis), rev0 and rev1s (results);
    - README and ATTRIBUTION separating original work from the GPL-3.0 magnetless_motor design and other sources;
    - GPL-3.0 licence.
- **Measured resistance:** 1.87–1.90 Ω on every phase of both stators. Ratings rev. B followed (1.89 Ω, 0.84 A pk).
- **Connection:** winding ends are brought out for star or delta, but the tests will be star only. The simulations correspond to star (i~A~ + i~B~ + i~C~ = 0). In delta, the 5.5 % third harmonic would drive about 40 mA rms circulating current.
- **Part C report:** 15 pages plus references, 11 figures and 8 IEEE references verified on Crossref. Three Crossref look-ups were refused or returned nothing.
    - The inductance was **corrected to about 25 µH per stator**. The Neumann filament sum over-predicted the single-board self-inductance about 4× against Maxwell's 2.07 µH, so PWM ripple is about 4× the earlier estimate.
- **Evening, rev. C after an external review.**
    - The ratings sheet became the "Predicted Operating Points": 0.84 A pk is a candidate thermal test point, and 1000 rpm the target test speed.
    - **Rev1-S parallel reworked:** S2's 13 % higher back-EMF drives 0.06–0.07 A pk circulating current and unequal sharing (S1 0.91 / S2 0.77 A pk at 1.68 A pk), so the terminal current was limited to 1.54 A pk (45.5 mN·m).
    - The mechanical-loss method was corrected to no-load I~q~, spin-down or a torque sensor.
    - Drive guidance became checks.
    - The measured 5.1 mm stack was recorded; it matches the model, which removes the −9 % risk.
    - Both technical reports were updated to 1.89 Ω (Rev0 at 1000 rpm and 1 A pk: 47.2 % efficiency). The Part C report and repository were updated and committed.

Where it stands on 19 October

Part C is complete for the modelled geometry. The as-built remodel waits on the measured air gaps.

Quantity (modelled geometry, 1000 rpm)	Rev0	Rev1-S (series)	Basis
K (T = 1.5·K·Î)	18.53 mN·m/A	39.50 mN·m/A	Rev0 full-motor FEM (job 1470376); Rev1-S board-by-board, equivalent-coil within 0.5 %
Torque per A pk	27.8 mN·m	59.2 mN·m	
Back-EMF, line	2.38 V rms	5.07 V rms	Rotating model, scaled to K
THD, phase / line	5.5 % / 0.7 %	5.2 % / 0.6 %	
Torque ripple / cogging	1.26 % / < 0.8 mN·m	1.16 % / < 0.8 mN·m	Cogging at solver noise
Backplate	B	max	0.91 T
Track eddy loss	20 mW	44 mW	Scales with speed²
Measured phase resistance	1.87–1.90 Ω	1.87–1.90 Ω per stator	Model 1.39 Ω; joints and leads make up the difference
Inductance	about 25 µH	about 50 µH	Corrected 19 Oct; measure it
Predicted EM torque at 0.84 A pk	23.4 mN·m	49.8 mN·m	Candidate thermal test point, not a rating

Open items

- [ ] Measure the assembled air gaps and the S2-to-R1 clearance, then remodel with the board-by-board FEM and the calibrated equivalent-coil model (backplate 80/25 mm).
- [ ] Open-circuit back-EMF test: 2.38 V rms line (Rev0) and 5.07 V rms (Rev1-S series) at 1000 rpm; S1 and S2 separately at about 2.38 and 2.69 V rms.
- [ ] Measure the phase inductance and the PWM current ripple with the actual inverter.
- [ ] Heat run to set the continuous current.

- [ ] Not done: the Rev1-S full-motor static solve (lost to the quota), Rev2-S (estimate only, about 60.5 mN·m/A), Rev1-D rotating runs, and the backplate/gap check runs (profiles written, never deployed).

Record correction found while writing this log. The HPC logs show the Rev0 full-motor solve was **job 1470376 on 32 cores** (243.8 GB allocated, 109.7 GB peak). The Rev0 README and the Part C report had said job 1466609 on 40 cores. Both are corrected in the package.